

linear_reg.ipynb

C:\Users> mhamd > Desktop > badawey > linear_reg.ipynb > new_data = pd.DataFrame({'Experience_Years': [8]})

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base (Python 3.12.7)

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```
[1] import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

Python 2.3s

```
[2] data = {'Experience_Years': [1, 2, 3, 4, 5, 6, 7],
        'Salary': [3000, 4200, 5100, 6000, 6800, 8000, 8900]}
```

Python 0.0s

```
[4] df = pd.DataFrame(data)
```

Python 0.0s

```
[6] X = df[['Experience_Years']]
y = df['Salary']
```

Python 0.0s

```
[7] X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Python 0.0s

```
[8] X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

Python 0.0s

```
model = LinearRegression()
```

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base (Python 3.12.7)

```
[9] model = LinearRegression()
model.fit(X_train, y_train)
```

Python 0.0s

LinearRegression

```
[10] y_pred = model.predict(X_test)
```

Python 0.0s

```
[11] print("Mean Squared Error (MSE):", mean_squared_error(y_test, y_pred))
print("R2 Score (Accuracy):", r2_score(y_test, y_pred))
```

Python 0.0s

Mean Squared Error (MSE): 14400.0
R2 Score (Accuracy): 0.96

```
[12] new_data = pd.DataFrame({'Experience_Years': [8]})
predicted_salary = model.predict(new_data)
print(f"\nPredicted Salary for 8 years experience: {predicted_salary[0]}")
```

Python 0.0s

Predicted Salary for 8 years experience: 9840.0

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